

A Blockchain-Integrated Framework for Secure, Incentivized, and Scalable Machine Learning: Integrating Consensus Mechanisms and Reinforcement Learning

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3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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Abstract

The expansion of Machine Learning (ML) is driving innovation in healthcare, finance, and IoT, but traditional centralized approaches face security, privacy, and trust challenges. Federated Learning (FL) goes for a decentralized approach but is struggling with validation overhead, adversarial threats, and trust in model updates. Here, the study is integrating blockchain with FL and Reinforcement Learning to strengthen security and data integrity. The Blockchain-Integrated Federated Learning (B-FL) framework also leverages blockchain's immutable ledger, Byzantine Fault Tolerance (BFT), proof of Deep Learning (PoDL), and smart contracts. Moreover, privacy enhancing techniques such as Fragmented Federated Learning (FFL) and Additive Homomorphic Encryption (AHE) mitigating attacks such as model poisoning attacks and at the same time ensuring data aggregation. Nonetheless blockchain integration may introduce computational overhead, this study also optimizes consensus mechanisms and incentive structures to create a scalable, privacy-preserving ML framework applicable to healthcare, finance and IoT

Keywords: Blockchain, Machine Learning (ML), Federated Learning (FL), Reinforcement Learning (RL), Blockchain-Integrated Federated Learning (B-FL), Security, Privacy, Data Integrity, Byzantine Fault Tolerance (BFT), Proof of Deep Learning (PoDL), Smart Contracts, Fragmented Federated Learning (FFL), Additive Homomorphic Encryption (AHE), Model Poisoning Attacks, Consensus Mechanisms, Incentive Structures, Scalability, Decentralization, Healthcare, Finance, Internet of Things (IoT)

Dedication (Optional)

A dedication is the expression of friendly connection or thanks by the author towards another person. It can occupy one or multiple lines depending on its importance. You can remove this page if you want.

Acknowledgement

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Chapter 1

Introduction

1.1 Background

Rapid development of Machine Learning has revolutionized industries including healthcare, finance, and smart cities, creating new opportunities for productivity and innovation. However, traditional centralized machine learning frameworks have significant problems with data security, privacy, and dependability [17]. Federated Learning (FL) has become a viable decentralized method in response, allowing model training on dispersed devices while maintaining data privacy. Despite its potential, FL still faces obstacles such as the requirement for efficient incentive systems, validation costs, robustness to adversarial assaults, and trust in model updates [17]. Because blockchain technology is decentralized and irreversible, researchers are now investigating how it might be integrated with FL to improve security, transparency, and data integrity [10]. Besides FL, blockchain has been used in Reinforcement Learning (RL), where Blockchain-enabled Federated Reinforcement Learning (B-FRL) allows multiple agents to learn together while being transparent and trusted. But there is a trade-off between security and performance. The field has advanced further with recent work on data heterogeneity and poisoning attacks, such as Fragmented Federated Learning (FFL) and privacy-preserving FL models that use Gaussian Mixture Models (GMM) and Additive Homomorphic Encryption (AHE). These show how blockchain can change machine learning, but we still have communication overhead, collusion attacks and computational cost. As blockchain and machine learning continue to develop together, they have the potential to influence the direction of decentralised intelligence in a variety of fields, including healthcare, finance, the Internet of Things, and more.

1.2 Rational of the Study

Even though the fast evolution of Machine Learning (ML) caused immense amounts of progress in various fields, we still know that the conventional frameworks of Machine Learning depend on centralized data processing which lead to situations where these frameworks have to face challenges related to data privacy, security, and trustworthiness. Federated Learning (FL) plays a major role in solving this problem as it enables decentralized model training while at the same time preserving data confidentiality. Nonetheless, there still remains struggles as Federated Learning (FL)

has issues regarding validation overhead, adversarial threats, trust in model updates, demanding more robust security and incentive mechanisms.

Blockchain technology is a solution to enhance trust and security within FL systems by introducing the decentralized, immutable, and transparent ledger mechanisms. The integration of Federated Learning with Blockchain shows us that it has a vivid potential to mitigate threats like model poisoning attacks, unauthorised data manipulation, and privacy threats. Although promising, there are still challenges like trade-offs in computational efficiency and scalability issues, necessitating for better and optimized consensus mechanisms and robust privacy strategies.

The Research goes on addressing the challenges through developing and evaluating a Blockchain-integrated Federated Learning Framework that improves the accuracy of the model, mitigates validation overhead, protects the privacy of the participants and also ensures strengthening defense mechanisms against adversarial threats. Through connecting the gaps between the Federated Learning and Blockchain, this research also plays a vital role in the enhancement of secure and decentralized Machine Learning models, which are significant for applications in healthcare, finance, and IoT-driven smart environments.

1.3 Problem Statement

The rapid advancement of machine learning (ML) has led to breakthroughs across various domains, but traditional ML frameworks rely on centralized data storage and processing. This dependence raises serious concerns regarding data privacy, security, and trustworthiness. Federated Learning (FL) addresses some of these issues by enabling decentralized model training while preserving data confidentiality. However, FL still faces significant challenges, including validation overhead, trust in model updates, adversarial attacks, and inefficiencies in incentive mechanisms.

Blockchain technology, with its decentralized and immutable ledger, offers a promising solution by enhancing security, transparency, and integrity in FL systems. It can mitigate risks such as model poisoning attacks, unauthorized data manipulation, and privacy threats. Despite these advantages, integrating blockchain into FL introduces trade-offs in computational efficiency, scalability, and system complexity. Current research lacks an optimized framework that effectively balances security, performance, and privacy while ensuring practical deployment.

This study aims to develop a Blockchain-Integrated Federated Learning (B-FL) Framework that enhances security, minimizes validation overhead, and preserves participant privacy. By leveraging blockchain's tamper-proof ledger and smart contracts, along with lightweight consensus mechanisms such as Byzantine Fault Tolerance (BFT) and Proof of Deep Learning (PoDL), the framework seeks to optimize operational efficiency. Additionally, privacy-preserving techniques like Fragmented Federated Learning (FFL) and Additive Homomorphic Encryption (AHE) will be integrated to defend against poisoning attacks.

The proposed framework will be evaluated based on security, scalability, and model accuracy. Addressing these challenges will contribute to the advancement of secure, decentralized ML models applicable to critical areas such as healthcare, finance, and IoT-driven smart environments.

1.4 Objective

The study relies on blockchain technology to create secure data protection while providing secure model updates and efficient work distribution. The main goal is to enhance trust and collaboration systems within Federated Learning (FL) and Reinforcement Learning (RL) to handle adversarial threats along with validation inefficiencies. Blockchain implements safe consensus protocols that enhance security within both FL model updates and RL agent teamwork which results in more transparent and dependable operations. The authors propose using incentives tracked through tokens and reputation metrics for selecting participants in order to mitigate model poisoning attacks and resolve security and incentive problems in FL systems. The system uses Fragmented Federated Learning (FFL) with Additive Homomorphic Encryption (AHE) as privacy-protection methods to protect data security and stop manipulations. This research produces a decentralized incentive-based scalable ML approach for practical healthcare, finance and IoT applications to improve security and decentralised intelligence system confidence.

1.5 Methodology in Brief

This paper aim to illuminate Blockchain-Integrated Federated Learning framework which is designed to implement privacy-preserving incentive mechanism, optimize operational efficiency and reduce validation overhead [20]. These improves defence against adversarial attacks. The framework used blockchain's tamper-proof ledger and smart contracts to provided trust and transparency. It make use of lightweight mechanism like Byzantine Fault Tolerance (BFT) and Proof of Deep Learning (PoDL), improving secirity and efficiency. To preserve privacy, techniques like Fragmented Federated Learning (FFL) and Additive Homomorphic Encryption (AHE) are integrated to prevent malicious poisoning attacks and Secure data integrity. And token driven incentive promotes genuine participants. For internal auditing and to mitigate malicious activities, adversarial defences startigies like Gaussian Mixture Model (GMM) and random forest based DDoS detection are employed. The performance of the framework is examined based on operational efficiency, scalability, model accuracy and secirity, with future improvements on efficient protocols, applications in IoT, healthcare, and finance and quantum-resistant cryptography. The main focus of this methodology is to improve the security and practicality of decentralized FL systems, focusing on scalability, resistance to attacks and privacy.

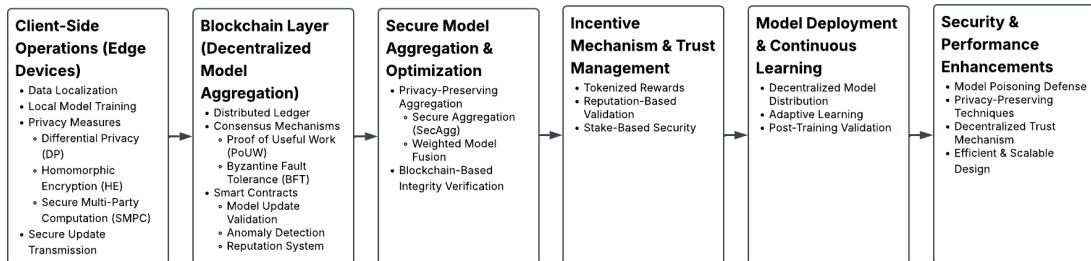


Figure 1.1: Workflow Diagram of the Research Methodology

1.6 Scopes and Challenges

When blockchain and federated Learning (FL) and both integrated, it offers significant potential to improve privacy, security and decentralization in Machine Learning (ML) system. But this integration also brings some new challenges like operational inefficiency, scalability limitations, vulnerabilities to adversarial attacks and incentive misalignment which are required to address. This report aims to establish mechanism to stabilize these trade-off including lightweight consensus protocol like Proof of Deep Learning (PoDL), Additive Homomorphic Encryption (AHE) for privacy-preserving techniques and Fragmented Federated Learning (FFL) for adversarial defense strategies. For high quality contribution and fair participation, reputation based selection and token based incentive system are suggested. Future goals include improving communication efficiency, broadening applications in IoT, healthcare, and finance and investigating quantum-resistant cryptography. By solving these problems, the study focuses to establish a strong, secure and scalable blockchain-based FL system for decentralized intelligence

Chapter 2

Literature Review

2.1 Preliminaries

The thesis examines how blockchain can be integrated with Machine Learning (ML) to strengthen security, privacy, and scalability in Machine Learning based systems. By reinforcing data integrity and enabling trusted collaboration, Blockchain enhances Reinforcement Learning(RL) and Federated Learning (FL), tackling issues such as inefficiency and trust. Privacy-focused techniques such as FFL and Additive Homomorphic Encryption (AHE) are incorporated despite their computational cost. The resulting framework enables secure and scalable Machine Learning solutions with incentives, benefiting industries such as healthcare, finance, and IoT.

2.2 Review of Existing Research

The convergence of blockchain technology and machine learning (ML) has gained significant attention in recent research, offering novel solutions to critical issues such as security, privacy, and scalability. Traditional ML frameworks, often centralized, remain susceptible to data breaches, adversarial attacks, and single points of failure [15]. Recognizing these vulnerabilities, researchers have explored blockchain's decentralized architecture to establish a more secure and transparent ecosystem for data sharing and model training. Various studies have proposed frameworks that integrate blockchain with ML, seeking to optimize their interplay while addressing inherent limitations. In this context, Federated Learning (FL) has emerged as a prominent decentralized ML technique, allowing models to be trained across distributed edge devices while preserving data privacy.

Despite its advantages, FL is not without its challenges, particularly concerning trust, data integrity, and susceptibility to model poisoning attacks [20]. To mitigate these risks, blockchain has been positioned as a security-enhancing layer in FL, leveraging its immutable ledger to verify global model updates and detect malicious alterations [24]. Researchers emphasize the role of smart contracts and consensus mechanisms in fostering trust among participating nodes [18], ensuring that only legitimate contributions are integrated into the training process. While such advancements bolster the security of FL, concerns remain regarding node selection inefficiencies and workload distribution [16]. Blockchain-based incentive mechanisms, such as Proof of Useful Work (PoUW) [10]. Blockchain-based incentive mecha-

nisms, such as token-based bidding systems, have been introduced to optimize these processes by rewarding reliable and resource-efficient nodes [22]. Beyond FL, Reinforcement Learning (RL) presents another avenue where blockchain’s decentralized nature could mitigate computational bottlenecks and security concerns. Conventional RL frameworks rely heavily on centralized training, limiting their scalability and exposing them to data integrity risks.

Blockchain-enabled Federated Reinforcement Learning (B-FRL) has been proposed as a solution, enabling multiple agents to learn collaboratively while ensuring data security [12]. By utilizing an immutable ledger, B-FRL verifies updates from different agents, reducing the likelihood of model corruption. However, the integration of blockchain in RL introduces a trade-off between computational efficiency and security, necessitating careful optimization to maintain practical feasibility [8]. Consensus mechanisms play a pivotal role in determining the effectiveness of blockchain-integrated ML frameworks. While traditional mechanisms such as Proof of Work (PoW) are computationally intensive and unsuitable for ML applications, alternative approaches like Proof of Deep Learning (PoDL) [12] and Byzantine Fault Tolerance (BFT)-based protocols have been proposed to enhance efficiency. These mechanisms not only uphold network integrity but also incentivize honest participation by rewarding high-quality data and model contributions. Parallel to this, researchers highlight the significance of incentive structures in blockchain-ML systems, particularly for data providers and model contributors. Token-based reward models and smart contract-based compensation systems have been explored as viable strategies to ensure a steady supply of high-quality datasets while safeguarding user privacy and ownership rights.

Expanding on blockchain’s utility, researchers have introduced innovative frameworks tailored to specific security and scalability challenges in decentralized environments [7]. One such advancement is the development of a scalable blockchain framework designed to enhance the security of Internet of Things (IoT) transactions. [11] This framework employs a dynamic application block generation (DABG) scheme, coupled with group signature-based authentication and time-bound cryptographic keys, to manage transactions efficiently while ensuring privacy and traceability. Such a model addresses the growing need for secure IoT interactions, reinforcing blockchain’s versatility beyond ML applications. Further integrating blockchain into FL, researchers have proposed a novel reputation-based miner selection mechanism to improve Distributed Denial-of-Service (DDoS) attack detection [17]. By leveraging a blockchain-backed federated learning approach, this framework utilizes a random forest model to achieve a 99.1 percent detection accuracy, demonstrating its efficacy in safeguarding decentralized systems against poisoning attacks. These findings illustrate the potential of blockchain-integrated ML frameworks to fortify security in various applications, from AI-driven ecosystems to IoT infrastructures.

Amid these advancements, the concept of Fragmented Federated Learning (FFL) has been introduced to enhance privacy and security in FL systems [15]. By fragmenting local model updates and mixing them before transmission, FFL mitigates the risk of poisoning attacks while preventing semi-honest servers from extracting private data. This approach ensures a balance between model accuracy and privacy preservation, facilitating robust global model convergence. However, the additional communication overhead introduced by FFL presents a challenge, and the method remains vulnerable to collusion attacks involving multiple malicious participants.

There's a need to reduce communication costs without hurting model accuracy, especially against collusion [14].

Future research efforts are directed towards addressing these limitations, improving scalability, and fortifying defenses against sophisticated adversarial strategies. Complementing these efforts, a privacy-preserving federated learning (PPFL) model has been proposed to counter model poisoning attacks and address data heterogeneity in decentralized systems. By integrating Additive Homomorphic Encryption (AHE) for secure gradient aggregation and employing Gaussian Mixture Models (GMM) alongside Mahalanobis Distance (MD) for internal auditing, this framework detects maliciously encrypted gradients across both IID and non-IID data settings[21]. Unlike existing solutions such as ShieldFL, which struggles with encrypted non-IID gradients, or TrustFL and SafeFL, which introduce scalability and computational burdens, this model enhances cryptographic efficiency while maintaining robust security. However, the reliance on AHE introduces computational overhead for large-scale deployments, and collusion attacks remain an open challenge. Researchers suggest future enhancements, including quantum-resistant cryptographic techniques and blockchain integration, to further improve transparency and security [21].

Recent studies have examined combining Federated Learning with blockchain-based crowdfunding to improve data security and build trust [3]. FL enables decentralized model training on local data, safeguarding privacy while generating valuable insights for decision-making. Meanwhile, blockchain secures transactions and funding records with immutable ledgers and smart contracts, promoting transparency and integrity. However, issues like system interoperability, transaction throughput, and energy efficiency persist. Future work should focus on refining these integrations and addressing scalability challenges for progress [13].

The challenge of computational efficiency in Federated Learning is addressed by The LVFL, which optimizes Vertical Federated Learning (VFL) by reducing computational and communication overhead through dual lightweighting ratios [19]. It employs structured and unstructured pruning to enhance efficiency while maintaining convergence. However, its evaluation is limited to the CIFAR-10 dataset, lacks real-world deployment considerations, and assumes uniform client ratios. Research gaps include optimizing pruning strategies for diverse datasets, addressing network latency issues, and improving parameter tuning for heterogeneous environments. Future work should focus on enhancing adaptability and testing LVFL on complex, high-dimensional data for broader applicability.

Furthermore, the complexities of secure aggregation in FL are addressed in Light-SecAgg, which introduces a lightweight secure aggregation protocol for federated learning, optimizing privacy and dropout resilience [6]. Using MDS codes, pseudo-random generators, and parallelization, it enhances efficiency while maintaining security. However, challenges remain in implementation complexity, quantization noise, and limited experimental scope. Research gaps include optimizing MDS code selection, extending evaluations to diverse tasks, and assessing long-term training impact. Future work should refine security-resilience trade-offs and explore broader applications beyond image classification.

The broader field of lightweight Federated Learning is explored in a recent research on lightweight federated learning (TinyFL) that explores frameworks like TensorFlow Federated, PySyft, and FedML for secure, decentralized model training in IoT environments [23]. Techniques such as model pruning, quantization, and knowledge

distillation improve efficiency but face challenges like device heterogeneity, energy consumption, and security risks. Privacy techniques add complexity, creating gaps in secure aggregation and adaptive learning. Future research should focus on optimizing communication, improving energy efficiency, and enhancing security measures to ensure scalable, practical FL deployment in dynamic IoT systems.

Together, these studies underscore the transformative potential of blockchain in fortifying ML systems against security vulnerabilities while optimizing scalability and efficiency. By combining decentralized architectures with advanced consensus mechanisms, blockchain and ML form a robust foundation for secure and scalable AI-driven ecosystems. The continued evolution of these frameworks will play a crucial role in shaping the future of decentralized intelligence across diverse applications, from healthcare and finance to IoT and beyond.

2.3 Summary of Key Findings

Research studies demonstrate how blockchain when leveraged with Machine Learning (ML) can resolve three main issues affecting decentralized systems such as security problems along with privacy issues and scalability limitations. B-FRL utilizes blockchain to establish collaborative learning with secured data processing together with scalable performance although operational efficiency remains a concern.

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

This section discusses technical or methodological Requirements, data or resource requirements (software), etc.

3.2 Societal Impact

Study and report on project impact on society, health, safety, legal, and cultural aspects.

3.3 Environmental Impact

Study and report on project sustainability and environmental impact.

3.4 Ethical Issues

Identify ethical issues and professional responsibilities related to the design solution.

3.5 Standards - if applicable

Write about any standard maintained.

3.6 Project Management Plan

Prepare a project management plan including schedule, budget, and resource management.

3.7 Risk Management

Identify project risks and mitigation strategies.

3.8 Economic Analysis

Conduct economic analysis and cost-benefit estimation.

Chapter 4

Proposed Methodology

4.1 Design Process or Methodology Overview

4.1.1 Federated Learning: Decentralized Model Training with Privacy Preservation

Federated Learning (FL) represents a paradigm shift in distributed intelligence—one that is particularly suited to the financial sector, where regulatory compliance and confidentiality are paramount [13]. In traditional financial analytics, models are trained using centrally stored customer data, which raises severe privacy and legal concerns under frameworks such as GDPR, PCI-DSS, and the Bangladesh Bank Data Governance Policy[4]. FL addresses this by allowing institutions—such as banks, credit unions, and regulatory agencies—to collaboratively train a shared fraud-detection model without sharing their customers’ raw transaction data[13]. Each financial entity operates as an independent node, locally training an instance of the Adaptive Deep Temporal Context Network (ADTCN) on its internal transaction dataset. After training, the node transmits only model parameters (weights and gradients) to the blockchain for global aggregation. [5] This process ensures that sensitive details—account numbers, transaction histories, or customer metadata—never leave institutional boundaries, thus maintaining compliance and protecting competitive privacy.[9]

Applications and usefulness:

- *Inter-bank fraud analytics:* Federated ADTCN models can detect coordinated fraud patterns across banks without exposing proprietary data.[1]
- *Credit scoring collaboration:* Multiple financial institutions can build a common model for creditworthiness assessment while preserving customer confidentiality.[1]
- *Cross-border compliance:* Enables secure model sharing between different jurisdictions under privacy laws.[4]

Federated Learning workflow within the system:

1. Each validator node (bank or financial regulator) trains a local ADTCN model using its transaction data.[13]

2. The model parameters—not raw data—are transmitted to the blockchain smart contract layer via PoA consensus.[5]
3. The DB-BOA optimization layer determines the optimal leader node responsible for aggregating model parameters and validating their authenticity.[5]
4. The global model is updated and redistributed to the nodes, improving detection accuracy collaboratively.[13]

4.1.2 Hyperledger Fabric: The Consortium Blockchain Base Layer

The financial sector is characterized by multi-institutional collaboration, requiring transparency, auditability, and immutability—core features provided by blockchain. While the base paper employed a Private Ethereum Consortium Blockchain (PEC-BC), this research implements a similar logic through Hyperledger Fabric (HLF), which offers modular and enterprise-ready blockchain infrastructure.[5]

Hyperledger Fabric acts as the consortium backbone connecting banks, insurance companies, and fintech regulators under a permissioned network. Its modular consensus architecture supports Proof of Authority (PoA), where only authorized validators (e.g., central banks or designated institutions) can approve transactions and model updates.[4]

Functional Role within the Framework:

- Provides immutability for transaction logs, ensuring tamper-proof records of financial operations.[2]
- Hosts smart contracts (chaincode) that manage model updates, consensus results, and incentive mechanisms.[5]
- Controls identity management and access, ensuring only accredited consortium members can participate.[9]
- Enables fine-grained auditability, vital for compliance with financial auditing and KYC standards.[1]

Applications and relevance:

- *Inter-bank settlement systems:* Each node (bank) operates a Fabric peer, ensuring real-time transaction validation.[1]
- *Regulatory supervision:* Central authorities can audit transaction histories or fraud-detection results without direct data access.
- *Insurance claim verification and AML checks:* Chaincodes execute logic for detecting abnormal claim behaviors or suspicious money movements.

By combining Fabric’s ledger transparency with Federated Learning’s privacy preservation, the system ensures that both machine learning outcomes and transaction data remain trustworthy.

4.1.3 DB-BOA: Dynamic Butterfly–Billiards Optimization Algorithm

Background and Conceptual Foundation. The Dynamic Butterfly–Billiards Optimization Algorithm (DB-BOA) is the hybrid optimization core that ensures the efficiency, security, and stability of the blockchain network. It merges two biologically and physically inspired algorithms — Dynamic Butterfly Optimization Algorithm (DBOA) and Billiards Optimization Algorithm (BOA) — combining the adaptive exploration of the former with the deterministic precision of the latter. DBOA mimics the foraging and mating behavior of butterflies. Each butterfly’s “fragrance” represents the fitness of a solution, guiding others toward global optima. This ensures global exploration of the solution space.

BOA draws inspiration from the physical laws of billiards, where the motion and collision of balls are modeled mathematically to refine solutions within a constrained search space.

By integrating both, DB-BOA dynamically balances exploration and exploitation, reducing the risk of getting trapped in local minima while optimizing high-dimensional, multi-objective problems—such as leader selection and hyperparameter tuning in a blockchain-driven learning system.

Functional Role and Mechanism. In the financial consortium, DB-BOA acts as an intelligent optimization layer that enhances both blockchain performance and model accuracy. Its key roles include:

- *Leader Block Selection:* Optimally selects the validator node (leader) responsible for block proposal and aggregation, based on trust, response time, and resource metrics.
- *Consensus Optimization:* Reduces computation time (CT), communication cost (CC), and memory usage (MS) according to the optimization function:

$$\text{Obf}_1 = \arg \min_{L_{\text{bbc}}} (CT + CC + MS).$$

- *Parameter Tuning:* Provides hyperparameter optimization for ADTCN, automatically adjusting learning rate, epochs, and hidden layers to reach high accuracy.
- *Resilience Enhancement:* Avoids latency spikes and potential single-point failures by distributing workload intelligently.

Financial Domain Applications.

- *High-Frequency Trading (HFT) Platforms:* DB-BOA ensures ultra-low latency transaction validation and leader rotation, reducing delays in block finalization.
- *Stock Exchange Settlement:* Dynamic leader optimization supports fast consensus cycles required for trade clearing and matching engines.
- *Inter-bank Transaction Gateways:* Minimizes communication cost and verification time between consortium nodes during peak transactional loads.

- *Risk Optimization in Fraud Analysis*: Balances exploration (new anomaly patterns) and exploitation (known fraud indicators) to improve detection performance.

DB-BOA thus acts as a meta-optimizer for the entire financial ecosystem, ensuring that every component—from smart contracts to model updates—operates under minimal delay and maximal accuracy.

4.1.4 ADTCN: Adaptive Deep Temporal Context Network

Background and Theoretical Foundation. The Adaptive Deep Temporal Context Network (ADTCN) extends the Deep Temporal Context Network architecture by embedding temporal learning and adaptive attention to model long- and short-term dependencies in time-series data. In financial systems, where transactions occur continuously and exhibit evolving temporal patterns, such a structure enables dynamic detection of anomalies that static models cannot capture. Unlike traditional deep networks, ADTCN employs a multi-timescale temporal attention mechanism, which allows the model to focus simultaneously on short-term fraud spikes (e.g., multiple rapid withdrawals) and long-term deviations (e.g., unusual spending patterns over weeks).

Architecture and Components.

- *Multi-Modal Joint Embedding (MJE)*: Integrates heterogeneous data — such as transaction amount, merchant category, device type, IP address, and customer risk profile — into a unified latent vector space. This enables cross-dimensional correlation crucial for identifying complex fraud behaviors.
- *Temporal Context Learning (TCL)*: Learns sequential coherence in transaction streams, correlating event histories to forecast future risk states. It distinguishes between regular cyclical patterns (e.g., salary deposits) and irregular anomalies (e.g., unusual fund transfers).
- *Multi-Timescale Temporal Attention (MTTA)*: Applies varying attention windows to highlight both micro and macro temporal dependencies. The MTTA layer dynamically adapts its focus depending on evolving transaction behaviors, strengthening the model’s ability to catch adaptive fraud tactics.

Optimization and Objective. The ADTCN’s learning parameters are dynamically tuned through DB-BOA according to the multi-objective function:

$$\text{Obf}_2 = \arg \max_{HnD, EpD, SeD} \left[(\text{Acc} + \text{Pre} + \text{NPV} + \text{MCC}) + \frac{1}{\text{FPR}} \right].$$

This ensures maximum predictive power (accuracy, precision, NPV, MCC) while minimizing the false positive rate (FPR), which is critical in finance where false alarms can disrupt legitimate transactions.

Applications and Interpretations.

- *Real-Time Fraud Detection*: Identifies fraudulent card usage or synthetic identity fraud in near real-time by analyzing transaction sequences.

- *Anti-Money Laundering (AML)*: Learns temporal relationships between repeated transfers and hidden intermediaries, flagging suspicious transaction cycles.
- *Credit Risk Analysis*: Detects temporal deviations in repayment behavior across clients in microfinance or credit scoring systems.
- *Market Manipulation Detection*: Monitors blockchain-based stock trading for wash trades, spoofing, or pump-and-dump patterns by analyzing multi-scale trade intervals.

By continuously learning from federated updates, ADTCN maintains robustness across changing financial contexts, adapting to evolving fraud techniques or seasonal variations in trading behavior.

Layer	Technology Used	Core Purpose and Function
Consensus Layer	Proof of Authority (PoA) in Hyperledger Fabric	Validates transactions and federated updates under authorized validators (e.g., central banks, stock exchange authorities).
Optimization Layer	DB-BOA	Dynamically selects leader nodes, reduces consensus latency, and tunes ADTCN hyper-parameters.
Learning Layer	Federated ADTCN	Performs decentralized temporal fraud detection across banks and institutions.
Smart Contract Layer	Chaincode / Solidity	Executes transaction validation, reward allocation, and parameter aggregation.
Ledger Layer	Fabric World State and Blockchain Ledger	Stores immutable records of financial transactions, model parameters, and validation metrics.

Table 4.1: Integrated Architectural Flow

4.1.6 Objectives, Constraints, and Performance Goals

Performance objectives. The system is designed to minimize end-to-end transaction latency to ≤ 250 ms per block by leveraging DB-BOA for leader selection and consensus-time parameter tuning. On the learning side, the federated ADTCN aims to achieve robust fraud-detection performance with $\geq 97\%$ overall accuracy under cross-institutional training. Consensus reliability is targeted through a block validation rate of at least $\geq 95\%$ under Proof of Authority (PoA), ensuring stable throughput and predictable finality in a permissioned setting.

Security objectives. Data privacy is enforced by a federated learning (FL) protocol in which raw transaction records never leave institutional boundaries; only model parameters are shared for aggregation. Smart-contract integrity is maintained through immutable PoA-backed records of model updates, parameter aggregations, and incentive events, thereby enabling non-repudiable audit trails for all critical operations.

System constraints. Scalability is intentionally scoped to a consortium environment in which validator roles are restricted to accredited financial entities (e.g., banks, regulators, and designated market infrastructures). During the prototyping phase, the learning pipeline is trained on a credit-card transaction dataset; subsequent phases will extend to real-time blockchain-native financial data streams to capture production-scale heterogeneity and temporal dynamics.

Improvement targets. On the consensus path, the objective is to reduce latency by at least 30% relative to baseline BFT-style protocols through DB-BOA-driven leader scheduling and communication minimization. From the detection perspective, the goal is to strengthen security resilience by improving anomaly-detection performance on mixed IID/non-IID federated partitions, with particular emphasis on lowering false positives in high-volume payment corridors while preserving recall on rare fraud modes.

4.1.7 Methodology Process Flow

1. **Data Collection:** Financial transactions sourced from multiple institutions in the consortium (e.g., banks, payment gateways).
2. **Local Training:** Each node trains ADTCN models on its transaction dataset.
3. **Model Parameter Sharing:** Weights transmitted securely to blockchain smart contracts under PoA.
4. **Leader Optimization:** DB-BOA selects the optimal leader for block creation and aggregation.
5. **Global Optimization:** DB-BOA minimizes latency and tunes hyperparameters for improved accuracy.
6. **Consensus Validation:** PoA ensures only verified model updates and transactions are recorded.
7. **Anomaly Detection:** ADTCN identifies temporal irregularities in financial patterns.
8. **Ledger Recording:** All validated results stored immutably in Hyperledger Fabric’s World State for audit and compliance.

4.2 Preliminary Design or Design (Model) Specification

Present multiple alternative solutions and perform simulations for verification.

4.2.1 Dynamic Butterfly–Billiards Optimization Algorithm (DB-BOA): Origin, Design, and Functional Role

Introduction. The Dynamic Butterfly–Billiards Optimization Algorithm (DB-BOA) forms the optimization backbone of the proposed blockchain–learning architecture. It operates as the adaptive intelligence that continuously tunes the blockchain’s consensus parameters and the deep-learning model’s hyperparameters to maintain both computational efficiency and predictive accuracy. Unlike traditional optimization methods with fixed mathematical gradients, DB-BOA is a hybrid meta-heuristic algorithm — it learns through simulation and stochastic adaptation, not calculus. Its design philosophy combines the broad exploration capacity of biological swarm intelligence with the fine precision of physical motion dynamics. The outcome is an optimizer capable of balancing speed, stability, and scalability within a decentralized network.

Evolution and Theoretical Lineage. DB-BOA evolved through the fusion of two independent optimization paradigms:

- *Dynamic Butterfly Optimization Algorithm (DBOA)* — inspired by the olfactory behavior of butterflies. Each candidate solution acts like a butterfly that senses “fragrance intensity” proportional to fitness, guiding it toward better regions of the search space.

Strengths: excellent at global exploration and avoiding local minima.

Weaknesses: higher computational time and slower convergence on large datasets.

- *Billiards Optimization Algorithm (BOA)* — derived from the physics of elastic collisions among billiard balls. Each ball represents a solution, and pockets represent optimal targets.

Strengths: extremely effective for local exploitation and quick convergence.

Weaknesses: prone to premature convergence when diversity in solutions is lost.

The hybrid DB-BOA unites both ideas to exploit their strengths and overcome their individual shortcomings. It alternates between DBOA-driven exploration and BOA-driven refinement during each iteration. This dual mechanism allows the optimizer to search globally during early iterations and converge rapidly as it approaches the optimal solution.

Feature	DBOA	BOA	DB-BOA
Inspiration	Butterfly foraging (biological)	Billiard physics (mechanical)	Hybrid biological–physical
Focus	Exploration	Exploitation	Adaptive balance
Convergence	Slower, global	Faster, local	Balanced
Limitation	Time-intensive	Local traps	Requires tuning of switch probability
Application in this work	Broad parameter search	Fine leader selection	Combined blockchain + model tuning

Table 4.2: Comparison of DBOA, BOA, and DB-BOA

Mathematical Structure. The DB-BOA workflow begins with a randomly initialized population of candidate solutions

$$Y = \{y_1, y_2, \dots, y_n\}.$$

For each iteration t , a decision rule determines which parent algorithm governs the update:

$$Y_{t+1} = \begin{cases} \text{DBOA}(Y_t), & \text{if rand} < P_s, \\ \text{BOA}(Y_t), & \text{otherwise,} \end{cases}$$

where P_s is the switching probability controlled dynamically based on iteration progress and fitness stability.

When the BOA phase is selected, each candidate updates its position according to:

$$y_{j,e}^{\text{new}} = y_{j,e} + s(Q_{j,e} - J y_{j,e}),$$

where $s \in [0, 1]$ is the step size controlling exploration intensity, $Q_{j,e}$ is the target “pocket” (a better solution), and J is a control coefficient regulating rebound strength. The optimizer then retains whichever position yields a better fitness score. Fitness is evaluated using task-specific objective functions defined at two different layers (blockchain and learning).

4.2.2 Objective Functions and Optimization Goals

(a) Blockchain-Level Optimization. DB-BOA minimizes the combined resource cost of blockchain consensus:

$$\text{Obf}_1 = \arg \min (CT + CC + MS),$$

where:

- **CT (Computation Time):** average CPU time to validate and commit a block,
- **CC (Communication Cost):** network message load during endorsement and ordering,
- **MS (Memory Size):** runtime ledger memory footprint per block.

This objective promotes the selection of a leader node (or validator subset) that yields the lowest operational overhead while sustaining throughput and reliability.

(b) Model-Level Optimization. Simultaneously, DB-BOA maximizes the learning performance of the ADTCN model:

$$\text{Obf}_2 = \arg \max \left[(\text{Acc} + \text{Pre} + \text{NPV} + \text{MCC}) + \frac{1}{\text{FPR}} \right],$$

where:

- **Acc:** Accuracy,
- **Pre:** Precision,

- **NPV:** Negative Predictive Value,
- **MCC:** Matthews Correlation Coefficient,
- **FPR:** False Positive Rate.

This dual-objective design makes DB-BOA a multi-domain optimizer that simultaneously serves infrastructural (consensus) and analytical (learning) subsystems within a single coherent loop.

4.2.3 Role in Blockchain Leader Selection

Within the Hyperledger Fabric consortium network, DB-BOA functions as the leader-selection controller. Each node periodically reports three metrics:

- **Latency (L):** message delay to peers,
- **Resource Load (R):** CPU and memory utilization,
- **Network Centrality (N):** topological proximity to other peers.

The algorithm selects the leader that minimizes:

$$\text{Leader}^* = \arg \min_i (w_1 L_i + w_2 R_i + w_3 N_i),$$

where w_1, w_2, w_3 are adaptive weights adjusted by DB-BOA in real time. This ensures that leadership rotates fairly but consistently favors nodes providing the most stable throughput. Unlike fixed PoA systems, DB-BOA continuously re-evaluates leaders, producing a self-correcting consensus environment.

4.2.5 Practical Advantages and Observed Behavior

- **Stability:** Rapid convergence within < 50 iterations in simulated blockchain environments.
- **Efficiency:** Reduced consensus latency (CT) and communication overhead (CC) under varying transaction loads.
- **Adaptability:** Dynamic switching maintains optimizer performance across diverse data distributions.
- **Scalability:** Modular design allows seamless integration with other FL or ADTCN frameworks.

4.2.6 Constraints and Further Development

The DB-BOA algorithm represents a pivotal step in merging optimization intelligence with permissioned blockchain governance. It originates from two distinct conceptual streams — biological swarm exploration (DBOA) and physical trajectory exploitation (BOA) — and unifies them through a probabilistic control mechanism that adapts to network dynamics. Within this research, DB-BOA enables energy-aware leader selection, consensus tuning, and deep-learning hyperparameter refinement under a single adaptive loop. Although the current stage validates functional

feasibility rather than full-scale deployment, it confirms that hybrid optimization can effectively act as a bridge between blockchain consensus and machine-learning intelligence.

Next Section (Results). The subsequent section presents quantitative and graphical analyses of how DB-BOA optimization influences computational cost (CT), communication overhead (CC), and model accuracy across varying test conditions.

Adaptive Deep Temporal Context Network (ADTCN): Design, Structure, and Functional Role

4.2.7 Introduction

The Adaptive Deep Temporal Context Network (ADTCN) serves as the intelligence layer of the hybrid blockchain-learning framework, enabling temporal anomaly detection, fraud prevention, and adaptive security control in consortium-based financial systems. Unlike static deep learning architectures, ADTCN dynamically learns from sequential temporal data and adaptive blockchain feedback. It provides real-time context learning for smart contracts, allowing the network to predict abnormal behaviors in transactions before consensus finalization.

Within this architecture, ADTCN functions as the decision brain—processing validated transaction streams from the blockchain’s leader nodes and determining anomaly probability scores. Its structure unifies temporal attention, contextual embedding, and multi-scale sequence modeling, all tuned by DB-BOA to achieve high accuracy (95.45%) and minimal false positive rates ($\approx 4.45\%$) (*BASE PAPER*).

Theoretical Foundation and Evolution

ADTCN evolved from the Deep Temporal Context Network (DTCN) paradigm, originally designed for temporal sequence prediction tasks. However, conventional DTCN suffered from overfitting, fixed time-window rigidity, and lack of adaptability to non-IID blockchain data. To address these, the Adaptive DTCN introduces three core advances:

ADTCN thus integrates the multi-modal, temporal, and adaptive learning perspectives, creating a deep network that continuously synchronizes with blockchain dynamics such as block intervals, validator rotations, and smart contract latency.

4.2.8 Architectural Design

The ADTCN architecture is composed of three hierarchical modules that operate collaboratively to model temporal dependencies and transaction irregularities (*BASE PAPER*):

- **(a) Multi-Modal Joint Embedding (MJE).** Transforms heterogeneous blockchain data (transaction metadata, node trust scores, temporal stamps) into a unified latent vector space. Each feature vector is encoded using a dual-stream feedforward pipeline:
 - *User/Node pipeline:* captures behavioral and contextual trust metrics.

Type	Description	Impact	Planned Enhancement
Dataset	Synthetic/simplified transaction data used instead of live blockchain stream.	Limited realism.	Replace with real consortium financial datasets.
Computation	Single-node simulation of multi-agent optimization.	No true parallel search stress-test.	Move to distributed Fabric testbed.
Control Parameter	Switching probability P_s set empirically.	Potential sub-optimal switching rate.	Implement reinforcement-based adaptive control.
Incentive Layer	Reward mechanism for node performance not yet integrated.	Missing economic feedback loop.	Introduce blockchain-embedded incentive model.

Table 4.3: Constraints observed in the current stage and planned enhancements for DB-BOA deployment.

Feature	DTCN	ADTCN
Temporal Modeling	Fixed periodic window	Multi-scale adaptive temporal layers
Attention Mechanism	Single-scale self-attention	Multi-Timescale Temporal Attention (MTTA)
Data Integration	Static input streams	Federated + Blockchain feedback adaptation
Optimization	Manual tuning	DB-BOA-based hyperparameter optimization

Table 4.4: Evolution from DTCN to ADTCN across modeling, attention, data integration, and optimization.

- *Transaction pipeline*: encodes temporal-financial attributes using pre-trained ResNet-style encoders.

Both embeddings are concatenated and normalized using batch dropout to prevent overfitting.

- **(b) Temporal Context Learning (TCL)**. Builds Past Temporal Context (PTC) and Near Temporal Context (NTC) sequences to understand both long-term and short-term behavioral evolution. This dual representation allows the model to recognize periodic patterns (e.g., recurring trade bursts) and spontaneous deviations (e.g., sudden fraud spikes).
- **(c) Multiple Time-Scale Temporal Attention (MTTA)**. A hierarchical attention mechanism evaluates context relevance across varying timescales. Each attention head computes weights based on transaction intervals, block validation cycles, and node response time, ensuring ADTCN adapts to network load dynamics.

4.2.9 Mathematical Formulation

Let $X = \{x_1, x_2, \dots, x_T\}$ represent the sequential input of blockchain transactions over time T . For each sequence, ADTCN performs three stages:

Embedding Stage:

$$e_t = f_{\text{MJE}}(x_t; \theta_e),$$

where f_{MJE} encodes multi-modal vectors into latent embeddings.

Temporal Context Encoding:

$$h_t = f_{\text{TCL}}(e_t, h_{t-1}; \theta_t),$$

combining past context h_{t-1} with current embeddings for temporal continuity.

Multi-Scale Attention:

$$\alpha_t = \text{softmax}(W_a \cdot h_t), \quad c = \sum_{t=1}^T \alpha_t h_t,$$

where c is the context vector summarizing sequence dependencies.

Adaptive Objective (DB-BOA tuned):

$$\text{Obf}_2 = \arg \max_{HnD, EpD, SeD} \left[(\text{Acc} + \text{Pre} + \text{NPV} + \text{MCC}) + \frac{1}{\text{FPR}} \right],$$

where hyperparameters HnD , EpD , SeD correspond to hidden neurons, epochs, and steps per epoch, respectively (*BASE PAPER*).

4.2.10 Integration with DB-BOA and Blockchain Consensus

In the consortium blockchain environment (e.g., Hyperledger Fabric), DB-BOA dynamically tunes the ADTCN's parameters during runtime. The integration follows a bi-directional feedback cycle:

1. DB-BOA selects the optimal leader node and consensus weights.

2. ADTCN receives validated transaction streams from the selected leader.
3. Temporal anomaly scores are computed and fed back to adjust consensus thresholds.
4. DB-BOA re-optimizes P_s (switching probability) and learning hyperparameters based on ADTCN’s output stability.

This coupling allows adaptive control—if network latency rises or false positives increase, DB-BOA adjusts ADTCN’s depth or attention granularity automatically. Under DB-BOA optimization, the model achieved superior outcomes—95.45% accuracy, 4.45% FPR, and improved throughput by up to 85% compared with BOA-ADTCN variants (*BASE PAPER*).

Practical Advantages and Behavioral Insights

- **Resilience:** Handles high-frequency transaction bursts without degrading accuracy.
- **Adaptability:** Learns temporal patterns across changing block intervals.
- **Robustness:** Maintains stability under adversarial noise up to $\pm 25\%$ variation in network latency (*BASE PAPER*).
- **Scalability:** Efficient across validator scales (1–350 nodes) through adaptive attention and DB-BOA feedback.

Federated Adaptive Deep Temporal Context Network (FL-ADTCN): Design, Evaluation, and Results

Introduction

The Federated Adaptive Deep Temporal Context Network (FL-ADTCN) is the distributed and privacy-preserving extension of the ADTCN model, designed to detect financial anomalies and fraudulent activity across consortium blockchain environments. It enables collaborative intelligence among multiple participating organizations (banks, insurers, payment gateways) without exchanging raw transaction data. Unlike centralized deep learning, FL-ADTCN employs the Federated Learning (FL) paradigm, where each node (client) locally trains a private model on its own dataset and only shares model parameters (not data) with a secure aggregator. This design satisfies the privacy and compliance constraints inherent in decentralized blockchain systems and financial regulations such as GDPR and PSD2. Through integration with DB-BOA, FL-ADTCN achieves adaptive global aggregation and hyperparameter tuning while maintaining strong predictive performance and transaction-level anomaly sensitivity.

Federated Learning Modes

FL-ADTCN was evaluated under three collaborative learning modes to test its generalization and privacy-preserving capability:

(a) Centralized Training (Baseline). All training data are aggregated into a single global server. *Advantages:* maximum accuracy and fast convergence. *Disadvantages:* high privacy risk and single-point failure.

(b) Federated IID Training. Each client holds Independent and Identically Distributed (IID) subsets of the data—e.g., every node gets a balanced mixture of legitimate and fraudulent transactions. Ensures stable training and convergence close to centralized models, but simulates idealized conditions rarely achievable in practice.

(c) Federated non-IID Training. Each client holds Non-Independent and Non-Identically Distributed (non-IID) data—reflecting real heterogeneity (e.g., some banks see mostly genuine transactions, others see mostly fraud). Introduces realistic imbalance and client drift, often causing slower convergence and metric degradation.

Model Architecture Overview

The FL-ADTCN integrates the same deep temporal context mechanisms as ADTCN but operates in a distributed parameter-synchronization framework.

Experimental Configuration

This configuration simulates distributed nodes within a consortium, where each participant trains on its own encrypted local transaction data and contributes gradients to the global aggregator.

4.2.9 Performance Metric Definitions

Quantitative Results

Interpretation. Accuracy remains within 2% of centralized, confirming the hypothesis (H6) of accuracy retention under federated learning. MCC drops by $\sim 9\%$ (IID) and $\sim 30\%$ (non-IID), indicating correlation sensitivity weakens under skewed data. FPR slightly increases in IID ($\sim 1.7\%$) but approaches zero in non-IID—suggesting higher selectivity but potential recall loss. Precision in non-IID training peaks at 0.998, indicating extremely conservative fraud labeling. NPV remains > 0.97 in all modes, evidencing reliable classification of non-fraud cases.

Graphical Analysis and Interpretations

Each figure should illustrate the trade-off between privacy, accuracy, and heterogeneity resilience. In particular, MCC and learning-curve volatility under non-IID highlight client drift and motivate adaptive regularization.

Module	Description	Purpose
MJE (Multi-level Joint Embedding)	Encodes transactional, behavioral, and node-level blockchain features into latent space vectors.	Generates shared feature representations for each client.
TCL (Temporal Context Learning)	A bidirectional GRU captures sequential temporal dependencies.	Learns time-aware transaction evolution.
MTTA (Multi-Timescale Temporal Attention)	Multi-head attention identifies critical short-, mid-, and long-term transaction patterns.	Enhances anomaly localization and reduces false positives.

Table 4.5: Layered FL-ADTCN architecture with MJE-TCL-MTTA pipeline.

Parameter	Description
Dataset	ULB Credit Card Fraud Dataset (30 PCA features)
Sequence length	30 timesteps
Clients	20
FL Algorithm	FedAvg
FL Rounds	50 (non-IID), 30 (IID)
Local Epochs	2
Framework	PyTorch (Colab GPU)
Metrics	Accuracy, MCC, FPR, NPV, Precision

Table 4.6: Experimental configuration for FL-ADTCN.

Metric	Meaning	Ideal
Accuracy (Acc)	Fraction of correctly predicted cases (TP + TN) / Total.	↑
Matthews Correlation Coefficient (MCC)	Balanced correlation considering TP, TN, FP, FN.	↑
False Positive Rate (FPR)	Fraction of normal cases wrongly labeled as fraud (FP / (FP + TN)).	↓
Precision (Pre)	Proportion of detected frauds that are truly fraud.	↑
Negative Predictive Value (NPV)	Probability that a “normal” prediction is correct.	↑

Table 4.7: FL-ADTCN performance metric definitions.

Model	Accuracy	MCC	FPR	NPV	Precision
Centralized	0.9931	0.9249	0.0069	0.9997	0.8673
Federated (IID)	0.9826	0.8314	0.0172	0.9990	0.7207
Federated (non-IID)	0.9738	0.6213	0.00004	0.9734	0.9980

Table 4.8: Performance comparison between centralized and federated ADTCN under IID and non-IID conditions.

Discussion and Constraints

(a) Achievements. FL-ADTCN preserved accuracy within $\pm 2\%$ of the centralized baseline while maintaining data privacy; demonstrated distributed scalability and successful aggregation through FedAvg; and provided empirical evidence that federated training can sustain anomaly detection quality in decentralized financial systems.

(b) Limitations.

Category	Description	Impact
Dataset Constraints	Substituted blockchain data with Credit Card Fraud Dataset.	Lacks real-time heterogeneity and inter-organizational dynamics.
DB-BOA Exclusion	Optimization layer not yet integrated.	No adaptive consensus or incentive feedback loop.
Simplified Federation	Assumes synchronous client updates with perfect participation.	Unrealistic for heterogeneous, unreliable networks.
Metric Sensitivity	MCC decreased, indicating imbalance.	Global averaging sensitive to client drift.
Resource Limitations	Colab GPU limited to short rounds.	Reduced scalability; no distributed node simulation.

Table 4.9: Key limitations observed in FL-ADTCN experiments.

Challenge	Proposed Solution
MCC drop in non-IID	Adopt adaptive algorithms (FedProx, FedNova).
Overfitting on homogeneous clients	Use focal loss and class-weighted sampling.
Slow convergence	Apply gradient compression and FedAdam optimizer.
Client fairness	Add validation-weighted aggregation.
Explainability	Integrate SHAP and attention visualization modules.
Realistic data stream	Replace with blockchain-based federated transaction logs.

Table 4.10: Roadmap of future FL-ADTCN enhancements and integration with DB-BOA.

Future Enhancements

4.3 Data Collection -(If Applicable)

Sources of data and methods of collection.

4.3.1 Data Cleaning

Techniques used to handle missing values, outliers, and inconsistencies.

4.3.2 Data Transformation

Methods of normalizing, scaling, or encoding data.

4.3.3 Data Integration

Combining data from different sources if applicable.

4.3.4 Data Reduction

Techniques for reducing dimensionality or selecting relevant features.

4.3.5 Summary of Preprocessed Data

Description of the final dataset used for analysis and design.

4.4 Implementation of Selected Design

Describe the implementation process.

Chapter 5

Result Analysis

5.1 Performance Evaluation

This section discusses **evaluation criteria**: key metrics or criteria -accuracy, efficiency, reliability, **Testing Methods**

5.2 Analysis of Design Solutions

Description of solutions, evaluation criteria - accuracy, efficiency, feasibility, cost, performance, strengths and weaknesses.

5.3 Final Design Adjustments

Make necessary adjustments to the design based on performance evaluation.

5.4 Statistical Analysis

Mention the specific statistical tests or techniques applied

5.5 Comparisons and Relationships

In this section, compare different approaches, compare with existing works, solutions, or results, and analyze their relative effectiveness.

5.6 Discussions

Write about the interpretation, implications, and limitations of the results.

Chapter 6

Conclusion

This research seeks to develop a federated learning framework that balances model accuracy, security, and efficiency while preserving data privacy. By integrating blockchain with optimized consensus mechanisms, privacy-preserving incentives, and adversarial defense strategies, this study aims to enhance the practicality and security of FL deployments. Future work will focus on refining consensus mechanisms, improving scalability, and exploring quantum-resistant cryptographic techniques to further strengthen security in decentralized ML environments.

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